



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	QUANTUM MECHANICS I				
Type of Course	Core in Major				
Semester	V				
Academic Level	300 - 399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	Fundamental Mathematics Concepts: Vector, Matrix, 2nd Order ODE, Probability.				
Course Summary	This comprehensive course aims to provide students with a solid foundation in quantum mechanics, delving into theoretical concepts, honing problem-solving skills, and offering exciting possibilities through hands-on simulations.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Demonstrate a deep understanding of the foundational principles of quantum mechanics	Understanding	Conceptual Knowledge	Written exams, quizzes
CO2	Solve complex quantum mechanical problems using mathematical formalism such as the Schrödinger equation	Applying	Procedural Knowledge	Problem sets, simulations
CO3	Analyze the quantum behavior of systems with discrete and continuous spectra	Analyzing	Conceptual Knowledge	Homework assignments, exams

CO4	Explain the physical significance of quantum mechanical operators and their properties	Understanding	Conceptual Knowledge	Class discussions, presentations
CO5	Predict the outcomes of quantum experiments and interpret their results within the framework of quantum theory	Evaluating	Conceptual Knowledge	Virtual lab experiments, projects
CO6	Apply quantum mechanics principles to understand topics such as box problem and quantum harmonic oscillator	Applying	Procedural Knowledge	Research papers, presentations
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48 +12)	Marks (70)
I	THE WAVE FUNCTION		6	10
	1	The Schrödinger Equation, The Statistical Interpretation.	1	
	2	Probability: Discrete Variables, Continuous Variables.	2	
	3	Normalization	1	
	4	Momentum	1	
	5	The Uncertainty Principle.	1	
	Sections 1.1 – 1.6 of chapter 1 of Book 1			
II	TIME – INDEPENDENT SCHRÖDINGER EQUATION		13	20
	6	Stationary States – Time-independent Schrodinger Equation, Time-independence of expectation values, A definite value of the total energy, General solution as a linear combination of separable solutions, Probability amplitudes.	3	
	7	The Infinite Square Well	3	
	8	The Free Particle – Wave packet, Phase and Group velocities	2	
	Sections 2.1, 2.2 and 2.4 of chapter 2 of Book 1			

	9	The Conservation of Probability: Probability Density, Probability Current Density, Interpretation of Equation of Continuity.	1	
	10	The Potential Step : Case $E > V_0$, Case $E < V_0$.	2	
	11	The Potential Barrier : Case $E > V_0$, Case $E < V_0$ – Tunneling.	2	
	Sections 3.6.4 of chapter 3, and sections 4.4, 4.5.1 and 4.5.2 of chapter 4 of Book 2			
III	MATHEMATICAL TOOLS OF QUANTUM MECHANICS		20	25
	12	The Hilbert Space and Wave Functions: The Linear Vector Space, The Hilbert Space, Dimension and Basis of a Vector Space, Square-Integrable Functions – Wave Functions	3	
	13	Dirac Notation: Kets, Bras, Bra-Kets and their Properties	2	
	14	Operators: General Definitions, Hermitian Adjoint and Its Properties, Hermitian Operators.	2	
	15	Commutator Algebra	1	
	16	Uncertainty Relation between Two Operators – General Relation and Heisenberg Uncertainty Relations	1	
	17	Functions of Operators	1	
	18	Eigenvalues and Eigenvectors of an Operator, Theorems 2.1 – 2.5.	2	
	19	Representation of Discrete Bases: Matrix Representation of Kets and Bras, Matrix Representation of Operators, Change of Bases and Unitary Transformations, Matrix Representation of the Eigenvalue Problem.	3	
	20	Representation of Continuous Basis: General Treatment, Position Representation, Momentum Representation, Connecting Position and Momentum Representations.	4	
	21	Matrix and Wave Mechanics: Matrix Mechanics, Wave Mechanics	1	
	Sections 2.2, 2.3, 2.4.1, 2.4.2, 2.4.4 – 2.4.6, 2.4.8, 2.5.1.1, 2.5.1.2, 2.5.2, 2.5.3, 2.6.1 – 2.6.4.3 and 2.7 of chapter 2 of Book 2			
IV	THE QUANTUM HARMONIC OSCILLATOR		9	15
	22	The Harmonic Oscillator: Energy Eigenvalues, Energy Eigenstates, Energy Eigenstates in Position Space (up to the first excited state only), The Matrix Representation of Various Operators, Expectation Values of Various Operators.	6	
	23	3D Problems in Cartesian Coordinates – General Treatment:	1	

		Separation of Variables		
	24	The Box Potential – Rectangular and Cubic Box Potentials, Degeneracy	1	
	25	The 3D Harmonic Oscillator: Anisotropic and Isotropic Oscillators, Degeneracy	1	
	Sections 4.8 – 4.8.5 of chapter 4, and sections 6.2.1, 6.2.3 and 6.2.4 of chapter 6 of Book 2			
Solved and unsolved problems of the relevant sections from the prescribed texts shall be discussed or given as assignment.				
V	OPEN – ENDED MODULE: COMPUTER SIMULATIONS OF QUANTUM SYSTEMS		12	
	Computer Simulations of quantum systems such as potential well, harmonic oscillator etc. can be done using appropriate numerical techniques and eigenvalue solvers in Python. The objectives can be to determine the energies of these systems and plot the probabilities of the states.			
Books and References:				
1. Introduction to Quantum Mechanics (Third Edition) by David J Griffiths (Book 1)				
2. Quantum Mechanics: Concepts and Applications (Second Edition) by Nouredine Zettili (Book 2)				
3. Principles of Quantum Mechanics by Ramamurti Shankar				
4. Quantum Mechanics: Theory and Applications" by Ajoy Ghatak and S. Lokanathan				
5. Lectures on Quantum Mechanics by B. K. Agarwal				
6. Quantum Mechanics: Non-Relativistic Theory" by L. D. Landau and E. M. Lifshitz				
7. NPTEL video lectures: https://nptel.ac.in/courses/122106034				

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO 4	PSO 5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	0	0	0	2	0	3	1	0	0	1	1	1
CO 2	3	2	3	0	3	0	3	1	1	0	1	1	1
CO 3	3	0	3	0	3	0	3	1	0	0	1	1	1
CO 4	3	0	0	0	2	0	3	1	0	0	1	1	1
CO 5	3	0	3	0	3	0	3	1	0	0	1	1	1
CO 6	3	0	3	0	3	0	3	1	0	0	3	1	0

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory / Practical Exam
- Assignments / Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory / Practical Exam	Assignment / Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	